

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Fundamentals of Data Science COURSE CODE:
DSA04

COURSE OUTCOME COVERED

CO1: Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)

QUESTIONS: 20

TotalMarks:100

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I M.Saron Jenish Raja(192324043) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Short Answer Test

1. A college wants to identify students who may fail in the final exam. How can Data Science help?

Ans: Through predictive analytics: models built on attendance, internal marks and past results predict at-risk students early so the college can give extra support.

2. A hospital has patient records for many years. How can Data Science be applied?

Ans: Analyse historical records to predict diseases, find treatment patterns, improve diagnosis and plan resources.

3. An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?

Ans: Recommendation systems (using customers' browsing and purchase history).

4. A bank wants to detect suspicious transactions. How can Data Science be useful?

Ans: Fraud detection: anomaly detection and machine learning flag unusual transaction patterns in real time.

5. A company wants to reduce customer loss. How can Data Science support this?

Ans: Customer churn prediction: identifies customers likely to leave so the company can target them with retention offers.

6. A school wants to improve student performance. Mention one benefit of Data Science in this case.

Ans: It identifies weak areas and learning patterns, enabling personalised teaching and timely intervention.

7. A supermarket wants to know which products sell more during festivals. How can Data Science help?

Ans: Sales trend and seasonal demand analysis (market basket analysis and forecasting) shows which products to stock more.

8. A transport department wants to reduce traffic congestion. How can Data Science be applied?

Ans: Analyse traffic and GPS sensor data to predict congestion, optimise signal timing and suggest alternate routes.

9. A farmer wants to improve crop yield. Give one use of Data Science.

Ans: Precision agriculture: analysing soil, weather and crop data to predict yield and guide irrigation and fertiliser use.

10. A company has data in Excel tables, customer reviews, images, and videos. Which facet of data is shown here?

Ans: Variety (structured, semi-structured and unstructured data).

11. A weather app updates temperature every minute. Which facet of data is involved?

Ans: Velocity (speed of data generation, real-time/streaming data).

12. A university stores student details in rows and columns. What type of data structure is this?

Ans: Structured data (tabular/relational).

13. Customer comments on social media are collected for analysis. What type of data is this?

Ans: Unstructured data (text).

14. A company collects a very large amount of daily user activity data from its mobile app. Which Big Data characteristic is applied?

Ans: Volume.

15. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?

Ans: Incomplete data (missing values) and inaccurate data (incorrect values), i.e. poor data quality/dirty data.

16. A business wants to convert customer data into useful sales decisions. Which Big Data characteristic is important?

Ans: Value.

17. A company stores large files in Hadoop Distributed File System. Which part of the Big Data ecosystem is used?

Ans: The storage layer (HDFS).

18. A company wants to process large datasets using parallel computation. Which Big Data tool can be used?

Ans: Hadoop MapReduce (or Apache Spark).

19. An organization wants to collect live sensor data from IoT devices. Which Big Data ecosystem component is suitable?

Ans: Apache Kafka (or Flume) for real-time data ingestion and streaming.

20. A company wants to visualize Big Data results for decision-makers. Which tool or method can be applied?

Ans: Data visualization tools such as Tableau or Power BI (dashboards and charts).